Saransh Gupta

Experienced AI Engineer with 4+ years of expertise in delivering AI/ML-driven solutions across enterprise environments. Proven track record as a researcher, with a strong focus on applying machine learning to real-world problems, driving innovation, and contributing to strategic initiatives that align with organizational goals.

saransh.official.iitkgp@gmail.com +91- 9530277421 LinkedIn Profile

Executive Profile

- Dedicated AI Engineer offering more than 4 years of experience in developing and productionizing Agentic Solutions, machine learning models and transforming data science prototypes into production grade solutions.
- Assessed strategies and validate modifications in machine learning models to enhance NLP systems continually, ensuring consistent improvement.
- Understanding of the concept of data science - advanced analytics, predictive modeler, machine learning algorithm in multiple technical and functional domains.
- Expertise in conducting full lifecycle analysis including data gathering and cleansing, deep dive advanced statistical analysis/modeling and recommendations to optimize performance.
- Skilled in leveraging Python, PyTorch, Transformers, BERT, scikit-learn, and TensorFlow to develop advanced machine learning models and optimize NLP systems.
- A focused individual with a zeal to learn and adapt to new technologies quickly; capabilities in managing critical situation.

Education & Certifications

- Indian Institute of Technology Kharagpur (2017 – 2022)
  B. Tech + M. Tech in Engineering Product Design, Industrial and Systems Engineering
  Grade: 8.09 / 10
- AWS Certified Cloud Practitioner (verify)

Publications

- "A hybrid approach for Anomaly prediction in server performance using clustering, outlier detection and gradient boosting", DevCon 2025, American Express
- "ClotCatcher: A Novel Natural Language Model to Accurately Adjudicate Venous Thromboembolism from Radiology Reports" BMC Medical Informatics and Decision Making doi: 10.1186/s12911-023-02369-z
- S. Gupta et al. "Integrative Network Modeling Highlights the Crucial Roles of Rho-GDI Signaling Pathway in the Progression of Non-Small Cell Lung Cancer" in IEEE - JBHI, 2022, doi: 10.1109/JBHI.2022.3190038
- Entity-aware Question-Answer Extraction for Shopping Guidance, Amazon Machine Learning Conference

Achievements

- Conferred with an exceptional performance rating at the American Express for impactful contribution to the organization in year 2024
- Received scholarship of 248 USD for Harvard College Project for Asian International Relations conference - 2022
- Featured as one of the Top 30 Undergraduate Achievers of IIT Kharagpur in the UG Achievers Directory 2020
- Awarded scholarship of 2200€ by The A*Midex Foundation of Aix-Marseille University, France, Feb 2020
- Selected among Top 5 percent out of all for the summer fellowship at Institute of Science Technology Austria
- Featured in the ISE Newsletter Autumn-2020 under Department Spotlight of ISE fights COVID- 19, 2020

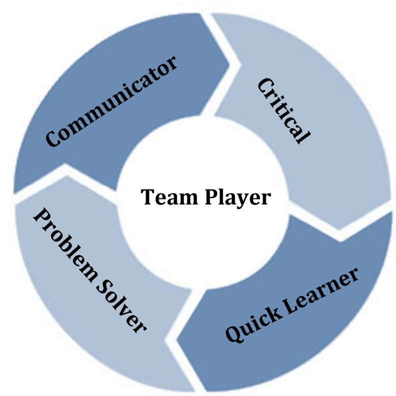
Links

- GitHub
- Google Scholar
- Website

Core Competencies

Natural Language Processing	Machine Learning Algorithms
Data Visualization	Agentic AI
Predictive Modeling	Amazon Web Services
Model Context Protocol	Elastic DB
Deep Learning	Neural Networks

Soft Skills



Technical Skills

- Python
- PyTorch
- Transformers
- Large Language Models
- Model Context Protocol
- LangGraph
- FastAPI
- scikit-learn
- TensorFlow
- SQL
- GCP
- AWS

Professional Experience

American Express | AI Engineer-II | April 2024 – Present

Project: Enterprise Developer Copilot for Internal Knowledge Retrieval

- Built a **reactive agent-based developer Copilot (GPT-5.2)** leveraging **MCP tools**, reducing query resolution time by **70%**
- Deployed **MCP tools using FastMCP** to expose internal docs, chats, and services, enabling unified and scalable enterprise knowledge access
- Consolidated fragmented data into a centralized **Elasticsearch index**, improving retrieval latency by **35%** and coverage to **90%**
- Developed a reusable **MCP tool framework**, standardizing integrations and accelerating onboarding of new tools by **50%**

Project: Enterprise Major Incident Management Copilot for faster incident resolution

- Built a **Copilot for major incident bridges,** analyzing live conversations, incidents, and inputs to guide Engineers, reducing MTTR by **55%**
- Developed a cross-encoder based **re-ranking pipeline** fetching top similar incidents, improving relevant search results by 30%
- Modeled **application dependency graphs** across payment systems, enabling proactive remediation and reducing escalation cycles by **25%**
- Deployed the solution in **production**, improving SLA adherence by **20%** and significantly accelerating cross-team incident coordination

American Express | Engineer-III | Aug 2022 – April 2024

Project: Failure cause identification of applications on generated Incident for their automated resolve

- Implemented a Question-Answer based strategy on top of raw dataset to identify failure cause of applications
- Achieved F1 Score of 0.84 by fine tuning a pre-trained BERT based Question-Answering model

Tools and Software: Python, PyTorch, GPT-5.2, BERT, Pandas, FastAPI, streamlit, ServiceNow, FastMCP, Elastic DB, LangGraph, GCP

Amazon Development Centre India | Applied Scientist – Intern | Jan 2022 - June 2022

Project: Generate Pre-Curated Question Bank (PCQB) by Question and Answer extraction from articles

- Developed a Transformers-based two-step model for Question Generation followed by the answer extraction
- Scrapped Texts, People Also Ask (PAA) questions and answers using queries related to the E- Commerce domain
- Achieved a Perplexity score of 82.3 on Question Generation by fine-tuning a T5 model on the PAA dataset
- Attained F-1 score of 0.79 on the answer extraction task by fine-tuning the T5-large model on the PAA dataset
- Deployed the two-step model pipeline on the streamlit-based demo web application that accepts user input

Tools and Software: Python, PyTorch, Transfer Learning, PAA, T5 Model, BERT, streamlit

ZS Associates| Data Science Associate – Intern | Jan 2021 – June 2021

Project 1: Extract biomedical text dataset, identify entities, and classify if there exists a relation between entities

- Created a pipeline to extract texts from PubMed database, identifying entities using Selenium and PubTator
- Implemented Binary Classification rules, devised four labeling functions using bio-verbs, co- occurrence of entities
- Generated a training dataset utilizing the four labeling functions in Snorkel by applying the Weak Supervision
- Achieved F1 score of 0.88 on the test dataset in relation-classification by fine-tuning RoBERTa base model

Project 2: Identify the type of relationship between two entities if it exists from the result of the Project-1

- Created a new set of three labeling functions for relation-type identification by using the results of the project-1
- Attained F1 score of 0.83 on the test dataset using XGBoost Model followed by feature engineering

Tools and Software: Python, PyTorch, Transfer Learning, Medline-Plus API, PubTator, Selenium, Snorkel

Research Experience

Emory University | Volunteer Researcher, Atlanta, GA, USA (Remote) | Jul 2022 – Aug 2023

Project: Predict the type of Venous thromboembolism (VTE), from the medical diagnosis and clinical Impressions

- Reduced manual adjudication of dataset by 20 times using pegasus paraphrasing model on sample dataset
- Achieved F1 score of 0.97 in predicting the type of VTE on test dataset by fine-tuning a Bio- BERT model
- Improved F1 score on test dataset by 20 percent by deploying paraphrasing and Bio-BERT finetuning pipeline

Tools and Software: Python, PyTorch, Transfer Learning, pegasus model, BERT

Osaka University | Research Assistant, Ibaraki, Osaka, Japan (Remote) | Jan 2020 - Dec 2020

Project: Predict Non-Small Cell Lung Cancer (NSCLC) using Machine Learning, identify potential drug targets

- Extracted 412 essential genes out of 10,077 by applying Boruta Feature selection on gene expression dataset
- Obtained F-1 score of 1.0 on validation, 0.98 on test dataset by using the XGBoost model to predict NSCLC
- Predicted drug targets for the NSCLC by simulating a Bayesian Network Model on Rho-GDI signaling pathway
- Discovered methodology leads to an accurate treatment of the disease impacting 85% of the lung cancer

Tools and Software: Python, Pandas, NumPy, matplotlib, XGBoost, Bayesian Network, networkX

Competitions and Conferences

- DevCon, Annual Developers and Inventors conference – American Express – May 2025
- The Harvard Project for Asian and International relations (HPAIR) – Hong Kong (SAR) - Aug 2023
- Annual Amazon Machine Learning Conference (AMLC) – Bengaluru, Karnataka - Aug 2022
- 23rd World Business Dialogue, Creation Lab at Evonik - Cologne, Germany - Jun 2022
- Amazon ML Summer School 2021: Offered PPI - Jul 2021
- International Conference on Human Interaction Emerging Technologies - Aug 2020
- Young Data Scientists annual meetup at Kaggle - days, Dubai World Trade Centre - Mar 2020
- Winner at Databuzz 2020 conducted by DoMS, IIT Madras - Jan 2020